Functions & Graphs

ANSWER KEY

Section 1: Function Notation & Operations

1 If $f(x) = 4x - 5$ and $g(x) = 2x^2 + 1$, what is $(f + g)(3)$?

a. 24

b. 26 ✓

c. 28

d. 30

$f(3) = 4(3) - 5 = 7$. $g(3) = 2(3^2) + 1 = 18 + 1 = 19$. $(f + g)(3) = 7 + 19 = 26$.

2 If $f(x) = 3x + 2$, what is the expression for $f(2a - 1)$?

a. $6a - 1$ ✓

b. $6a + 1$

c. $6a - 3$

d. $5a + 1$

$f(2a - 1) = 3(2a - 1) + 2 = 6a - 3 + 2 = 6a - 1$.

3 What is the domain of the function $h(x) = \frac{\sqrt{x-4}}{x-7}$?

a. $[4, 7) \cup (7, \infty)$ ✓

b. $[4, \infty)$

c. $(4, 7) \cup (7, \infty)$

d. $x \neq 7$

For the square root, $x - 4 \geq 0 \Rightarrow x \geq 4$. Denominator cannot be zero: $x \neq 7$. Domain is $[4, 7) \cup (7, \infty)$.

Section 2: Composite Functions & Inverses

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4 If $f(x) = 2x - 3$ and $g(x) = x^2 + 4$, what is $(f \circ g)(2)$?

a. 13 ✓

b. 15

c. 17

d. 19

$g(2) = 2^2 + 4 = 8$. Then $f(g(2)) = f(8) = 2(8) - 3 = 16 - 3 = 13$.

5 What is the inverse function of $f(x) = \frac{5x+3}{2}$?

a. $f^{-1}(x) = \frac{2x-3}{5}$ ✓

b. $f^{-1}(x) = \frac{2x+3}{5}$

c. $f^{-1}(x) = \frac{5x-2}{3}$

d. $f^{-1}(x) = \frac{2}{5x+3}$

Let $y = \frac{5x+3}{2} \Rightarrow 2y = 5x + 3 \Rightarrow 5x = 2y - 3 \Rightarrow x = \frac{2y-3}{5}$. Swapping variables: $f^{-1}(x) = \frac{2x-3}{5}$.

6 If $f(f^{-1}(x)) = x$ for all x, and $f(4) = 9$, what is $f^{-1}(9)$?

a. 4 ✓

b. $\frac{1}{4}$

c. $\frac{1}{9}$

d. 9

By definition of the inverse function, if $f(a) = b$, then $f^{-1}(b) = a$. Since $f(4) = 9$, $f^{-1}(9) = 4$.

Section 3: Graph Transformations

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7 How is the graph of $y = f(x - 4) + 2$ obtained from the graph of $y = f(x)$?

a. **Shifted 4 units right and 2 units up ✓**

b. Shifted 4 units left and 2 units up

c. Shifted 4 units right and 2 units down

d. Shifted 4 units left and 2 units down

The transformation $x - 4$ shifts the graph right by 4, and $+2$ shifts the graph up by 2.

8 The graph of $y = x^2$ is vertically stretched by a factor of 3 and reflected across the x-axis. What is its equation?

a. $y = -3x^2$ ✓

b. $y = 3(-x)^2$

c. $y = -\frac{1}{3}x^2$

d. $y = -(3x)^2$

Vertical stretch by 3 multiplies the function by 3: $3x^2$. Reflection across the x-axis negates the function: $-3x^2$.

9 Which of the following functions has origin symmetry (odd function)?

a. $f(x) = x^3 - 4x$ ✓

b. $f(x) = x^2 + 5$

c. $f(x) = 3x - 2$

d. $f(x) = x^4 - x^2$

An odd function satisfies $f(-x) = -f(x)$. Here $(-x)^3 - 4(-x) = -x^3 + 4x = -(x^3 - 4x) = -f(x)$.

Section 4: Rational Graphs & Asymptotes

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10 What is the vertical asymptote of $f(x) = \frac{3x+1}{x-5}$?

a. $x = 5$ ✓

b. $x = -5$

c. $y = 3$

d. $y = 5$

A vertical asymptote occurs where the denominator is zero (and numerator is non-zero): $x - 5 = 0 \Rightarrow x = 5$.

11 What is the horizontal asymptote of $g(x) = \frac{4x^2-1}{2x^2+3}$?

a. $y = 2$ ✓

b. $y = 0$

c. $y = 4$

d. No horizontal asymptote

Degrees are equal. The horizontal asymptote is the ratio of leading coefficients: $y = \frac{4}{2} = 2$.

12 What are the coordinates of the hole in the graph of $h(x) = \frac{x^2-9}{x-3}$?

a. $(3, 6)$ ✓

b. $(3, 0)$

c. $(-3, 0)$

d. $(3, 3)$

Factor: $\frac{(x-3)(x+3)}{x-3} = x + 3$ for $x \neq 3$. At $x = 3$, $y = 3 + 3 = 6$. The hole is at $(3, 6)$.

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